

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM ID	SWTID-2026-7684
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	5 MARKS

CHAPTER – IV PROJECT DESIGN

4.1 – Problem Solution Fit

Problem Statement

Electricity consumption continues to increase due to population growth, industrial development, and expanding technological usage. Large amounts of consumption data are generated, but understanding usage patterns and converting data into actionable insights remains difficult.

Organizations and users often face challenges such as:

- Limited understanding of electricity consumption behaviour.
- Difficulty identifying trends and peak demand periods.
- Inefficient energy planning and resource allocation.
- Lack of clear visualization for decision-making.
- Challenges in forecasting future electricity requirements.

Proposed Solution

The project provides an interactive analytical platform that transforms electricity consumption data into meaningful visual insights and predictive information.

Key solutions include:

- Collect and organize electricity consumption data.
- Analyse historical consumption patterns.